National University of Computer and Emerging Sciences
Lahore Campus

Blockchain and Cryptocurrency (CS4049)

Date: February 24th 2026

Course Instructor(s)

Dr. Sayed Qaiser Ali Shah

Sessional-I Exam

Total Time (Hrs):	**1**
Total Marks:	**49**
Total Questions:	**5**

_______________ _______________ ______________________________

Roll No Section Student Signature

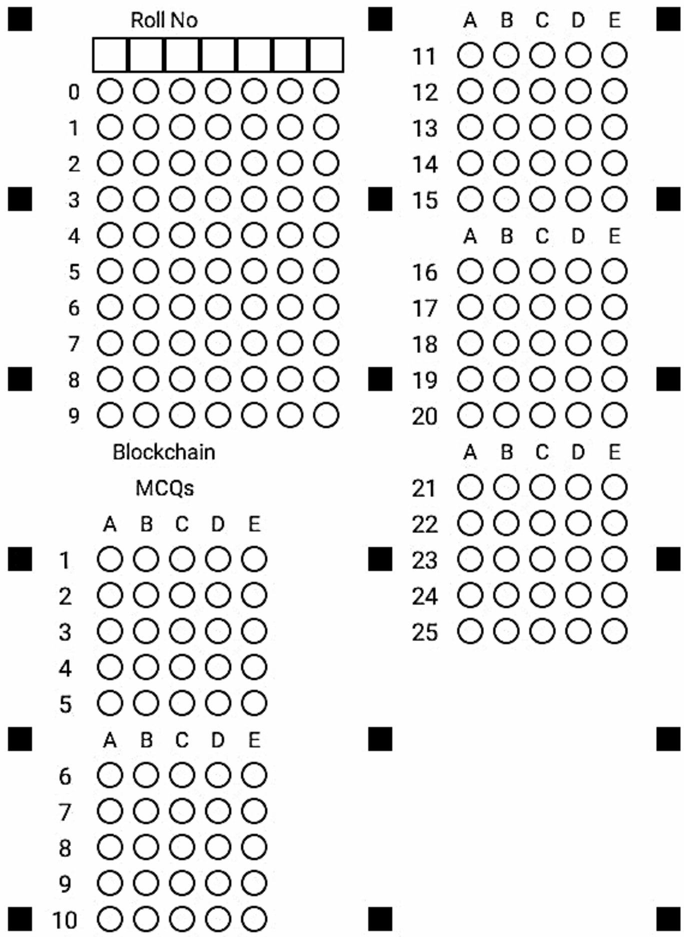
Do not write below this line

Attempt all the questions.

MCQs Answer Sheet

Instructions for Answering MCQs:

- Attempt MCQs on Bubble Sheet only otherwise will not be checked.
- Fill the Roll Number without dash in the provided field and replacing the character with respective digits (L=0). For Example: 22L-9980 => 2209980)
- Cutting, overwriting, multiple answers, or not correctly filling the circle would be considered incorrect and choose the most relevant options.



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[CLO 1: Identify key components behind the design of blockchain and to understand the types of applications that best fit the model of blockchain]

[Part-A]

Q1: Multiple Choice Questions (MCQs) **[25 marks]**

1. A block in a blockchain contains:
   a. Only transactions
   b. Transactions, hash, and metadata
   c. Only the hash of the previous block
   d. IP addresses of users
   e. Smart contract source code
2. The immutability of blockchain data is ensured by:
   a. Password protection
   b. Cryptographic hashing and linking blocks
   c. Anti-virus software
   d. Cloud storage
   e. Two-factor authentication
3. What happens if someone changes a previous block in the chain?
   a. Network ignores it
   b. Only that block is updated
   c. Hash mismatch invalidates the chain
   d. Wallets reset
   e. Nothing happens
4. Blockchain operates in which type of network?
   a. Peer-to-peer
   b. Ring topology
   c. Star topology
   d. Centralized client-server
   e. Mesh-based FTP
5. Bitcoin mining rewards reduce after every ______________ approximately:
   a. 2100 blocks
   b. 21000 blocks
   c. 10000 blocks
   d. 10 million blocks
   e. 210000 blocks
6. Which of the following stores your private keys?
   a. Node
   b. Wallet
   c. Miner
   d. Block
   e. API
7. The smallest unit of Ethereum is called:
   a. Ether
   b. Wei
   c. Microbitcoin
   d. Satoshi
   e. Bit

8. The Bitcoin blockchain is:
   a. Private
   b. Hybrid
   c. Public and permissionless
   d. Sidechain only
   e. Centralized
9. What happens when two miners mine a block at the same time?
   a. Both are accepted
   b. The chain splits forever
   c. Bitcoin stops functioning
   d. The longer chain eventually wins
   e. The last block is deleted What is a Bitcoin node?
10. A smart contract:
    a. Is manually executed
    b. Needs admin approval
    c. Executes based on conditions written in code
    d. Works like a PDF
    e. Is stored on USB
11. Which of the following best explains the role of the Merkle root in a Bitcoin block?
    a. It encrypts all transactions for privacy
    b. It stores the hash of the previous block
    c. It tracks the balance of all addresses
    d. It summarizes all transactions for efficient verification
    e. It controls the mining difficulty
12. What role does the difficulty target play in Bitcoin's Proof of Work system?
    a. It determines how many nodes are allowed to mine
    b. It adjusts transaction fees dynamically
    c. It regulates the average block mining time
    d. It defines the maximum allowed block size
    e. It encrypts block data
13. Alice is sending a Bitcoin transaction to Bob. She sets a very low transaction fee. What is the most likely outcome?
    a. The transaction is canceled by the sender
    b. The transaction is delayed or possibly ignored by miners
    c. The transaction is processed immediately
    d. The blockchain auto-adjusts the fee
    e. Bob must pay the rest of the fee
14. A developer writes a smart contract with a public function that transfers ETH without checking the sender's identity. What risk does this introduce?
    a. The contract will auto-delete
    b. Anyone could call the function and steal ETH
    c. It will fail due to missing ABI
    d. It uses too much gas
    e. Only the miner can access the function
15. What is the main purpose of a smart contract?
    a. To send emails
    b. To automate agreements
    c. To mine cryptocurrency
    d. To design websites
    e. To store images

16. A DApp developer deploys a contract but later finds a bug. They forgot to include an upgrade mechanism. What is their best option now?
    a. Edit the deployed code directly
    b. Use selfdestruct() to recover Ether
    c. Contact miners to revert the block
    d. Roll back the blockchain
    e. Redeploy a fixed version and redirect users manually
17. A user imports a wallet using a 12-word seed phrase on a new phone. What happens to their funds?
    a. The wallet creates a new private key
    b. The funds are lost permanently
    c. The wallet becomes invalid
    d. The address changes but the balance remains
    e. The same private key and funds are restored
18. A Bitcoin user tries to send coins from an address, but the wallet shows "insufficient inputs". What is the likely cause?
    a. The mempool is full
    b. The address uses Proof of Stake
    c. The required UTXOs are already spent
    d. The transaction fee is too high
    e. The transaction is too old
19. Smart contracts execute automatically when:
    a. The government approves them
    b. A lawyer signs them
    c. Predefined conditions are met
    d. The blockchain is restarted
    e. The miner requests execution
20. Which of the following is required to interact with a smart contract?
    a. Internet connection
    b. Blockchain network access
    c. Cryptocurrency wallet
    d. All of the above
    e. None of the above
21. Which of the following is NOT a feature of smart contracts?
    a. Automation
    b. Transparency
    c. Central control
    d. Immutability
    e. Deterministic execution
22. A smart contract can manage:
    a. Digital assets
    b. Voting systems
    c. Token transfers
    d. Escrow services
    e. All of the above
23. In Bitcoin, what does “6 confirmations” mean?
    a. Six miners approved the transaction
    b. The transaction appears in six different wallets
    c. Six blocks have been mined after the block containing the transaction
    d. Six nodes verified the transaction
    e. The transaction fee was multiplied by six
24. A transaction with 0 confirmations is also called:

a. Confirmed transaction
b. Coinbase transaction
c. Mempool transaction
d. Finalized transaction
e. Orphan transaction

25. A miner in Bitcoin is responsible for:
a. Verifying Transactions
b. creating blocks
c. Storing Users passwords
d. Setting exchange prices
e. Both A and B

Solution:

1. b 2. b 3. c 4. a 5. e 6. b 7. b 8. c 9. d 10. c 11. d 12. c 13. b 14. b 15. b

16. e 17. e 18. c 19. c 20. d 21. c 22. e 23. c 24. c 25. e

[Part-B]

Scenario: A group of Bitcoin miners begins censoring transactions from specific wallets due to political pressure. These transactions are not included in any blocks.

[CLO 2: Understand why blockchain is considered "secure" and "immutable" and the design of Bitcoin and other cryptocurrencies]
Q2: Can miners choose to exclude certain transactions? Why or why not? **[6 marks]**

Yes, **miners can technically choose to exclude certain transactions** from the blocks they produce, but this goes **against the decentralized and neutral ethos** of Bitcoin.

Explanation:

1. **Miner Autonomy:**
   Each miner independently selects which transactions to include in their block. They generally prioritize transactions offering the **highest fees** (in satoshis per byte).
   Hence, a miner *can* omit transactions from specific wallets.
2. **Protocol Perspective:**
   The **Bitcoin protocol** does **not enforce inclusion of all valid transactions**—only that transactions follow consensus rules (valid signatures, no double spend, etc.).
   Exclusion does not make a block invalid.
3. **Decentralization Counter-Force:**
   However, censorship is difficult to sustain because:
   - Other miners can include the censored transactions in their own blocks.
   - The longest-chain rule incentivizes inclusion of all profitable (fee-paying) transactions.
   - Users can increase fees to attract inclusion ("fee market pressure").
4. **Economic and Reputational Cost:**
   Persistent censorship lowers miners' profitability and may lead to **loss of trust** and **network forks** if the community views them as acting maliciously.

[CLO 2: Understand why blockchain is considered "secure" and "immutable" and the design of Bitcoin and other cryptocurrencies]
Q3: Propose and explain potential strategies that could reduce or prevent miner-driven transaction censorship in Proof of Work (PoW) blockchain networks. **[6 marks]**

Several mechanisms can reduce or neutralize censorship attempts by miners in Proof-of-Work (PoW) systems like Bitcoin:

1 Decentralization of Mining Power

- Encourage many independent miners and pool diversity to prevent cartel behavior.
- Discourage concentration through anti-pool protocols (e.g., Stratum V2 with job negotiation).

2 Stratum V2 Protocol

- Allows individual miners to choose transactions for inclusion rather than pool operators, preventing centralized control over block templates.

3 Fee-based Incentives

- Users can increase transaction fees to make censorship economically irrational.
- Higher-fee transactions attract miners who prioritize revenue.

4 Relay Networks and Mempool Diversity

- Use multiple relay networks (FIBRE, Falcon) to propagate transactions quickly so that censored transactions reach *non-censoring* miners.

5 "Inclusion Proof" Mechanisms

- Future Bitcoin research explores proof-of-inclusion or censorship-resistant relay nodes to track missing transactions.

6 Soft Governance / Social Layer

- The community can soft-fork or penalize identified censoring entities by changing mining pools or adopting new protocols.

Scenario: A large mining pool controlling 30% of the network hash power suddenly goes offline due to regulatory action.

[CLO 2: Understand why blockchain is considered "secure" and "immutable" and the design of Bitcoin and other cryptocurrencies]
Q4: How does Bitcoin's difficulty adjustment mechanism respond? **[6 marks]**

Bitcoin adjusts mining difficulty **every 2016 blocks** (approximately every 2 weeks under normal conditions).

Because blocks are now being mined more slowly:

- It will take **longer than two weeks** to mine the next 2016 blocks.
- When the adjustment occurs, the protocol automatically **reduces the mining difficulty**.
- The goal is to bring the average block time back to **10 minutes**.

The difficulty adjusts proportionally based on how long the previous 2016 blocks took to mine.

After adjustment:

- Block time stabilizes near 10 minutes again (assuming hash rate remains constant).

[CLO 2: Understand why blockchain is considered "secure" and "immutable" and the design of Bitcoin and other cryptocurrencies]

Q5: Will network security increase or decrease? Explain. **[6 marks]**

Short-term: Security decreases.

Reason:

- **Lower total hash power means less computational cost to attack the network.**
- **A malicious actor now needs fewer resources to perform attacks (e.g., double spending).**
- **The threshold for a 51% attack becomes easier to reach because total network hash rate is lower.**

Long-term:
If miners relocate and hash power returns, security restores. After difficulty adjustment, block timing stabilizes, but overall security depends on total hash power, not difficulty alone.
